SERVER: HPC



Breaking the Memory Capacity-Bandwidth Tradeoff for Agentic Inference

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Final V1 - 10/12/2026

Contributors : Amphenol, Crealien, Credo, Empower(Analog Devices), FIT, LOTES



Why Inference Keeps Eating Memory

More tokens drive bandwidth · bigger models + longer context drive capacity

1 **Bigger models — MoE**

10× total params, same active set

CAPACITY

2 **Long context + agents**

KV cache grows with every token

CAPACITY BANDWIDTH

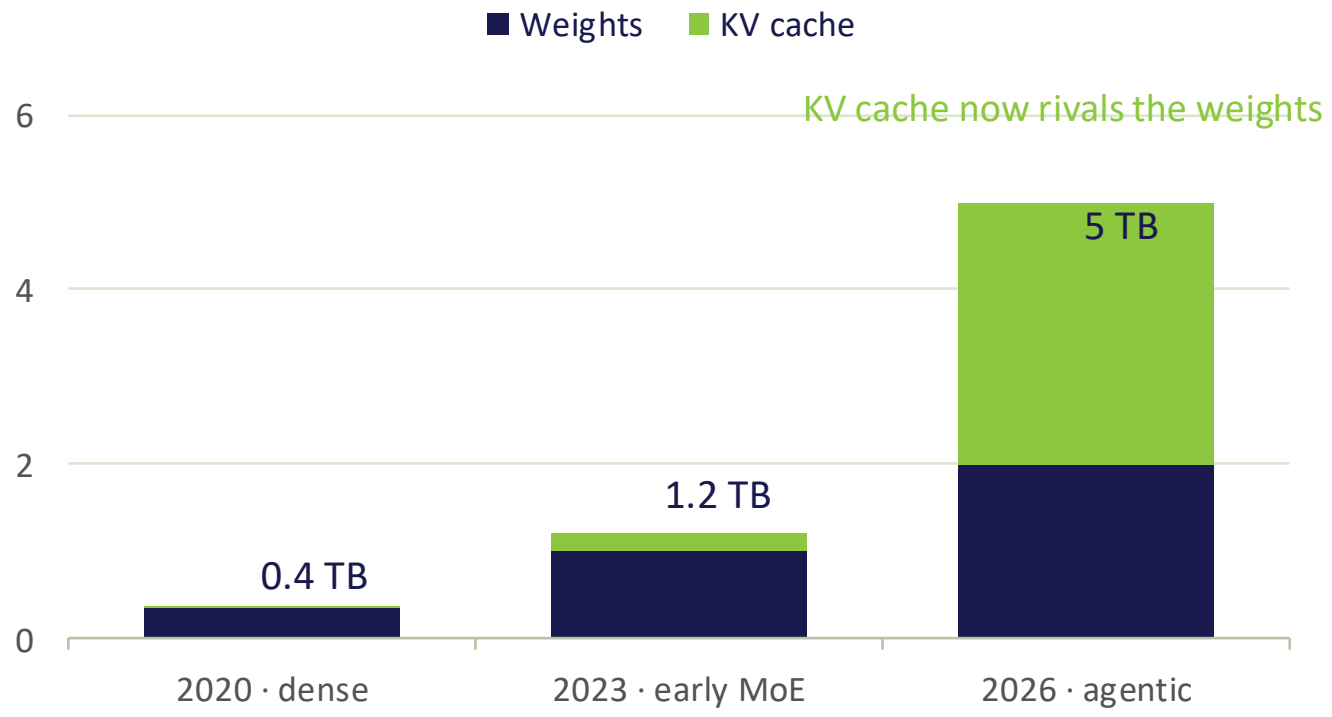
3 **Reasoning tokens**

10–100× tokens per answer

BANDWIDTH

Every token re-reads it all —
capacity drags bandwidth with it.

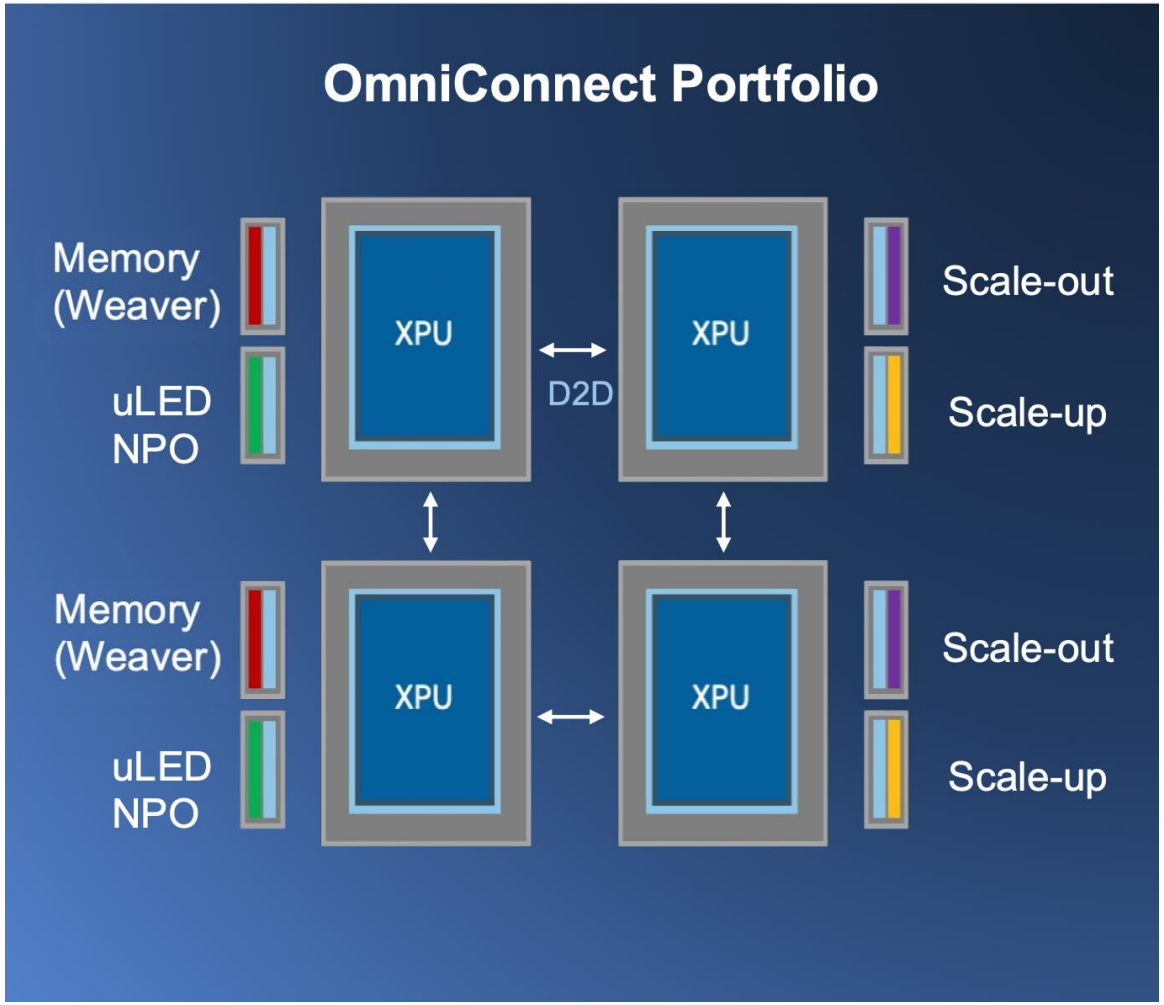
Memory per serving instance



Illustrative, order-of-magnitude

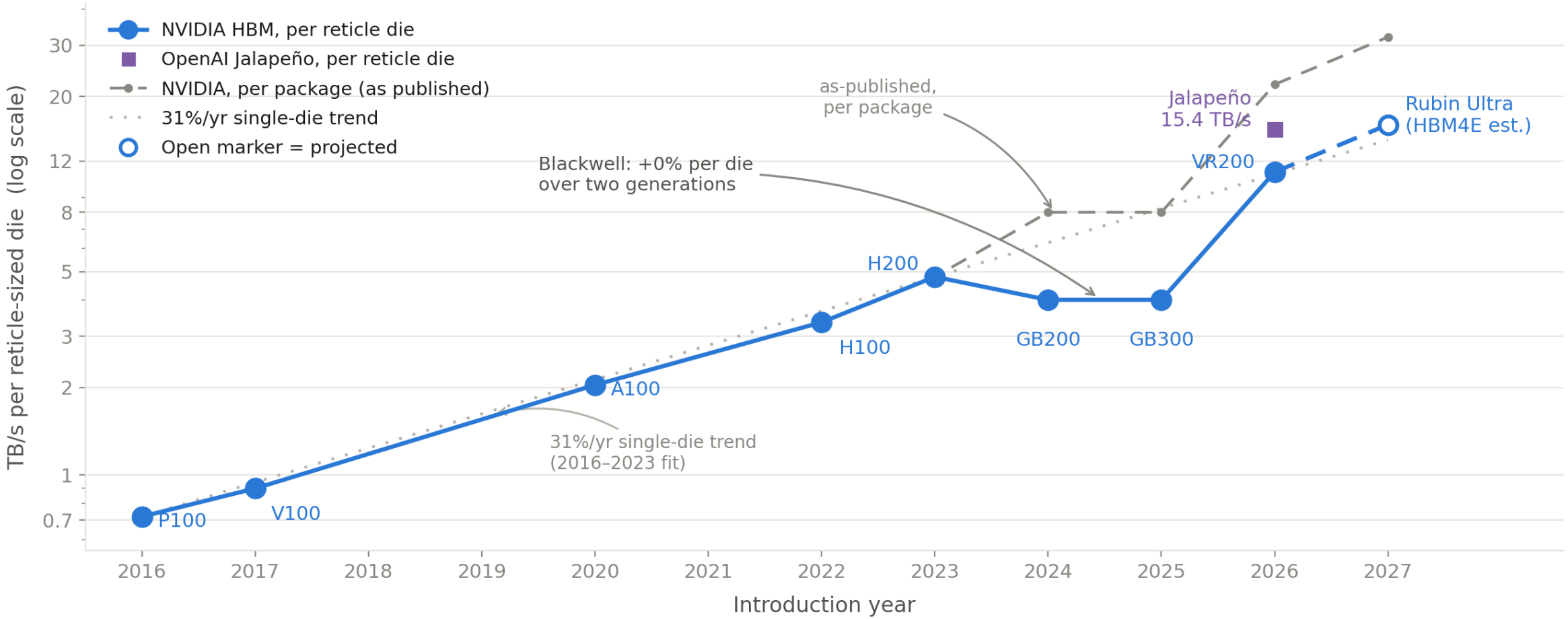
Credo's Lightweight Serial Interconnect, LSI, OCP Contribution

- OmniConnect LSI
  - Unified XPU IO Chiplet Interface
    - Late IO Binding, e.g. X% Memory, Y% Scale-up, Z% Scale-Out
  - AXI over VSR 112Gbps Serdes @ 1.15pJ/bit
  - Low Latency, ~35ns one way
- Compliant with OCP FCSA Section 5.12
  - Non-Coherent Memory Traffic Interface
- 600 Serdes East/West IO ~8 TBytes/s
  - Per Reticule Sized Die



Bandwidth per Reticle-Sized Die

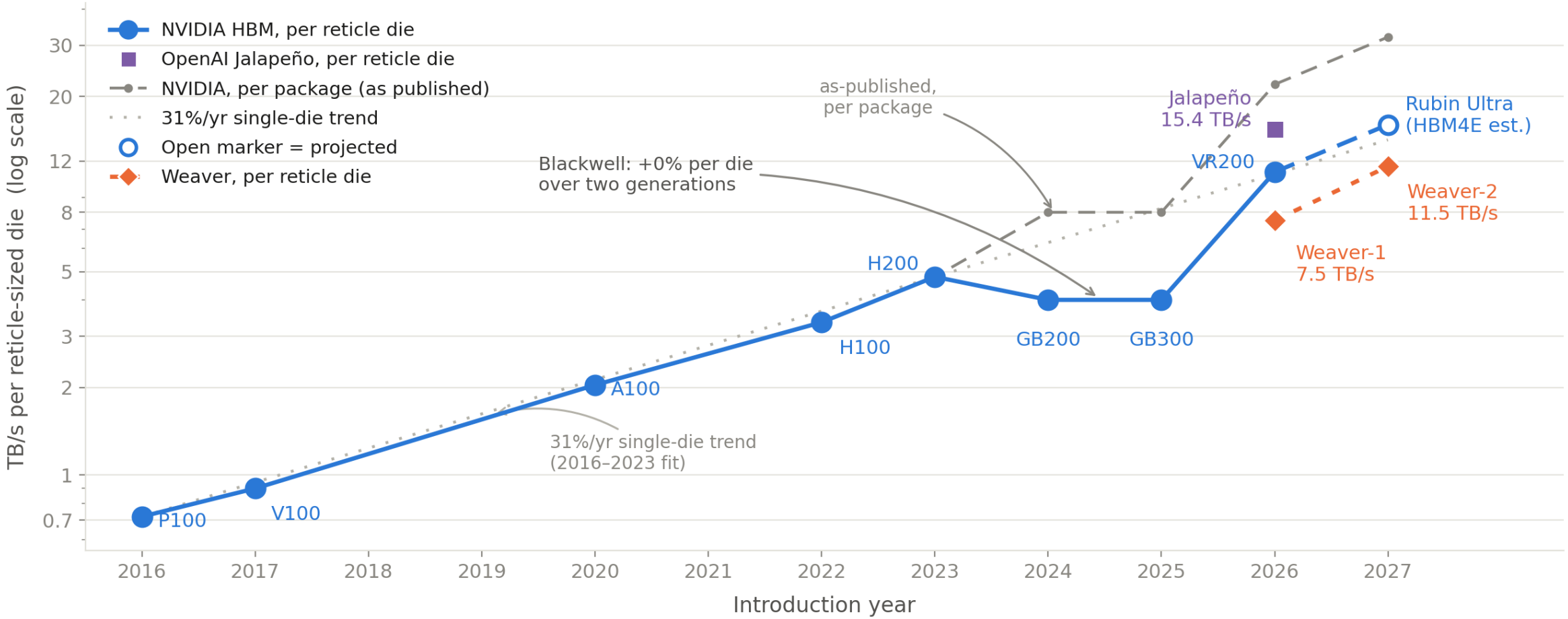
Jalapeño went back to one reticle die with six HBM4 stacks; Rubin Ultra needs early HBM4E to stay ahead



Package figures from SemiAnalysis. Rubin Ultra 2027 is the revised two-die / 8-stack part (four-die version cancelled, June 2026), plotted on Samsung HBM4E — 16 Gbps, 4.0 TB/s per stack, GTC 2026 — assuming early HBM4E adoption; a mainstream HBM4 8-Hi SKU is also reported at ~21 TB/s per package. Jalapeño from Hot Chips 2026 — 15.4 TB/s, 216 GiB HBM4, 700 W.

Bandwidth per Reticle-Sized Die

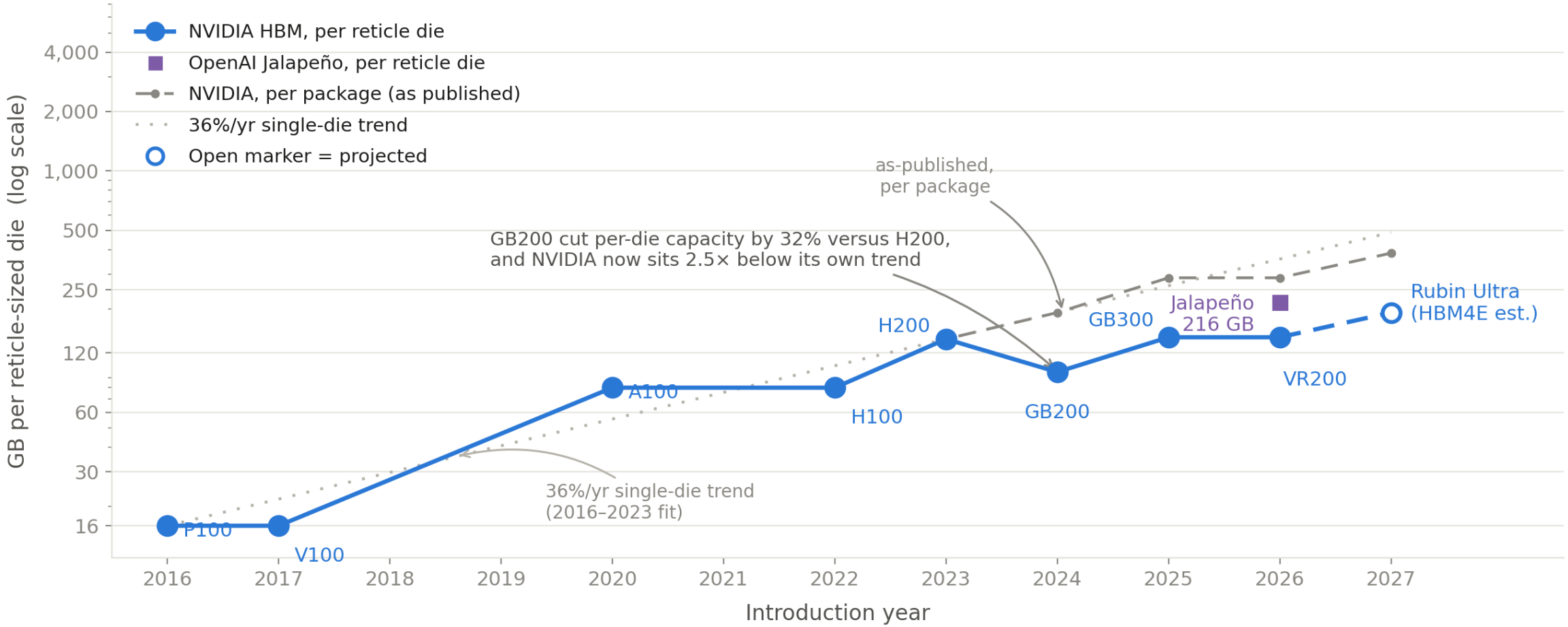
Weaver-2 reaches 11.5 TB/s per die — 72% of Rubin Ultra, with no HBM and no advanced packaging



Package figures from SemiAnalysis. Rubin Ultra 2027 is the revised two-die / 8-stack part (four-die version cancelled, June 2026), plotted on Samsung HBM4E — 16 Gbps, 4.0 TB/s per stack, GTC 2026 — assuming early HBM4E adoption; a mainstream HBM4 8-Hi SKU is also reported at ~21 TB/s per package. Jalapeño from Hot Chips 2026 — 15.4 TB/s, 216 GiB HBM4, 700 W. Weaver figures are Nallasway targets.

Capacity per Reticle-Sized Die

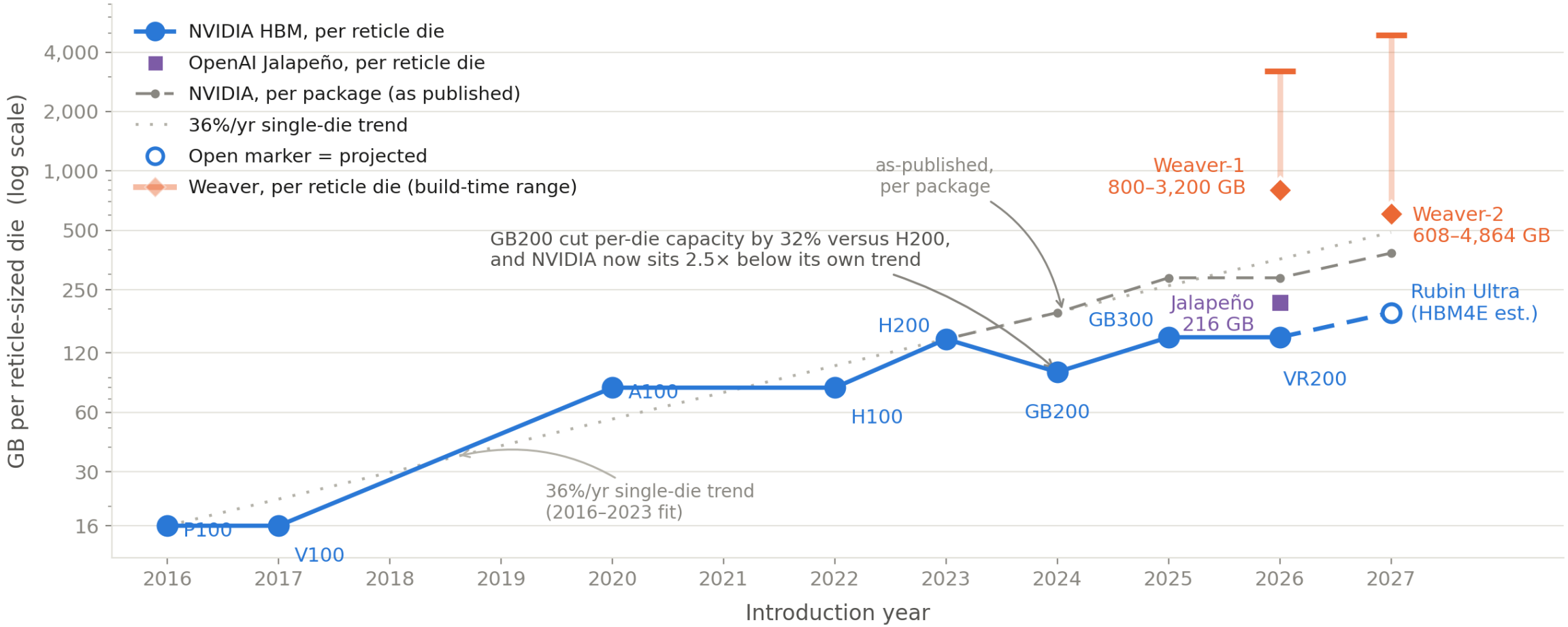
Per-die capacity growth broke at Blackwell — 36%/yr through 2023, 8%/yr since, and nobody is back on trend



Package figures from SemiAnalysis. Rubin Ultra 2027 on 12-high HBM4E — 48 GB per stack, four per reticle die — same early-HBM4E basis as the bandwidth slide; a mainstream HBM4 8-Hi SKU is also reported at 192 GB per package. Jalapeño: 216 GiB HBM4, six stacks, one die (Hot Chips 2026).

Capacity per Reticle-Sized Die

Weaver puts per-die capacity back above trend — 3.2× the HBM part at minimum, 25× at maximum

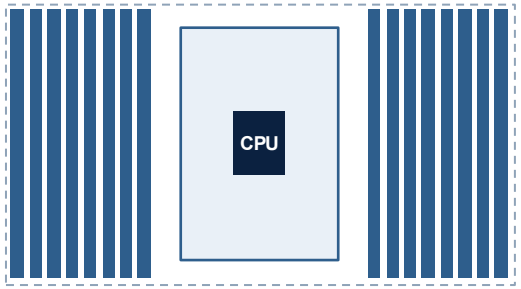


Package figures from SemiAnalysis. Rubin Ultra 2027 on 12-high HBM4E — 48 GB per stack, four per reticle die — same early-HBM4E basis as the bandwidth slide; a mainstream HBM4 8-Hi SKU is also reported at 192 GB per package. Jalapeño: 216 GiB HBM4, six stacks, one die (Hot Chips 2026). Weaver range is the build-time module choice — 50 × 16–64 GB, 76 × 8–32 GB.

Three ways to attach memory today

50 mm

CPU + 16 MRDIMMs
DDR5 MRDIMM-8800 · 256 GB per DIMM

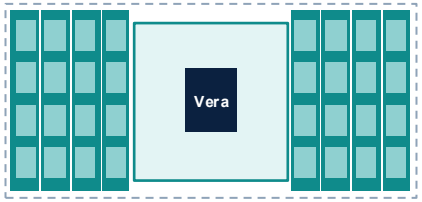


Bounding Area ≈ 330 cm²

4 TB CAPACITY
1.1 TB/s BANDWIDTH

Capacity, not bandwidth

Vera + 8 SOCAMMs
SOCAMM2 LPDDR5X-9600 · up to 48 GB each

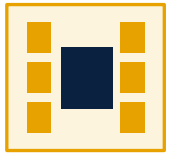


Bounding Area ≈ 178 cm²

1.5 TB CAPACITY
1.2 TB/s BANDWIDTH

Compact, but capped

Jalapeño + 6 HBM4
HBM4 · 36GB each



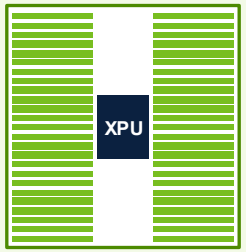
Bounding Area = 65 cm²

216 GB CAPACITY
15.4 TB/s BANDWIDTH

Bandwidth, not capacity

Reticule XPU + 50 OC-DDIMMs

LPDDR5X-9600 OC-DDIMM - 64GB per DIMM (OmniConnect Differential DIMMs)



Bounding Area = 138 cm²

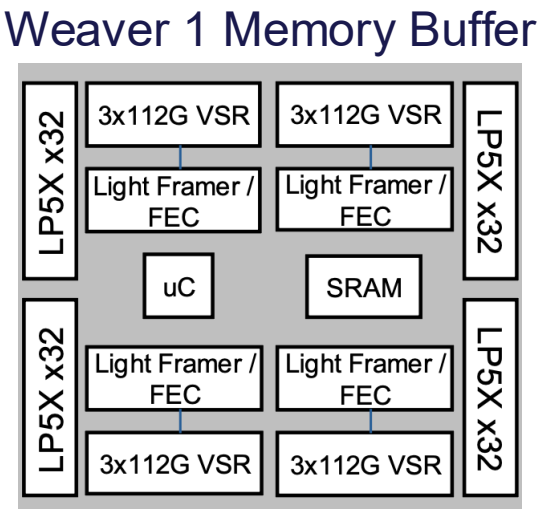
3.2 TB CAPACITY
7.5 TB/s BANDWIDTH

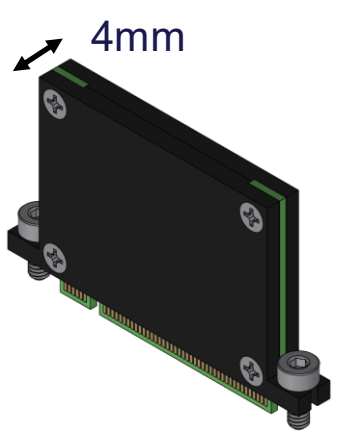
One Memory Tier

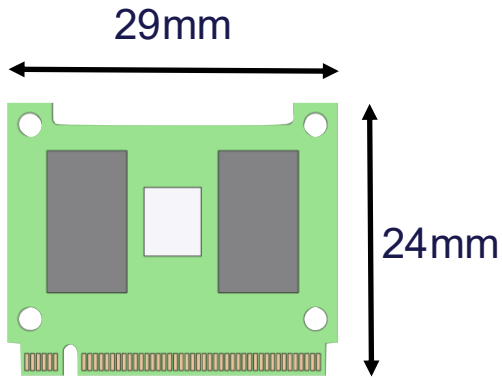
Delivers Compact Capacity at Bandwidth

OmniConnect Weaver 1 Memory Buffer & Module

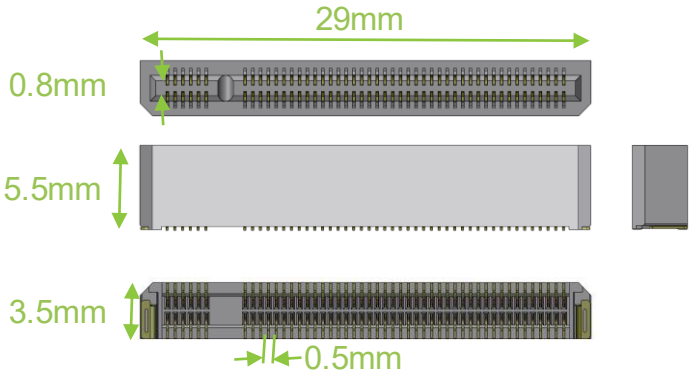
- Weaver 1 Memory Buffer
  - Supports 4 LPDDDR5X x32 Channels
    - 9.6Gbps Data Rate
  - 12x 112Gb Transceivers (3 per Channel)
- Proposed Weaver 1 OC-DDIMM, OmniConnect Differential DIMM
  - 2x LPDDR5X x16 4 Ch 563b packages
  - 16 GByte to 64 GByte Capacity
    - 4 to 8, 16Gb or 32Gb Die Stacks per package
- OC-DDIMM Connector
  - Based on M.2 Connector
  - 96 pads at 0.5mm pitch on a 0.8mm thick Substrate



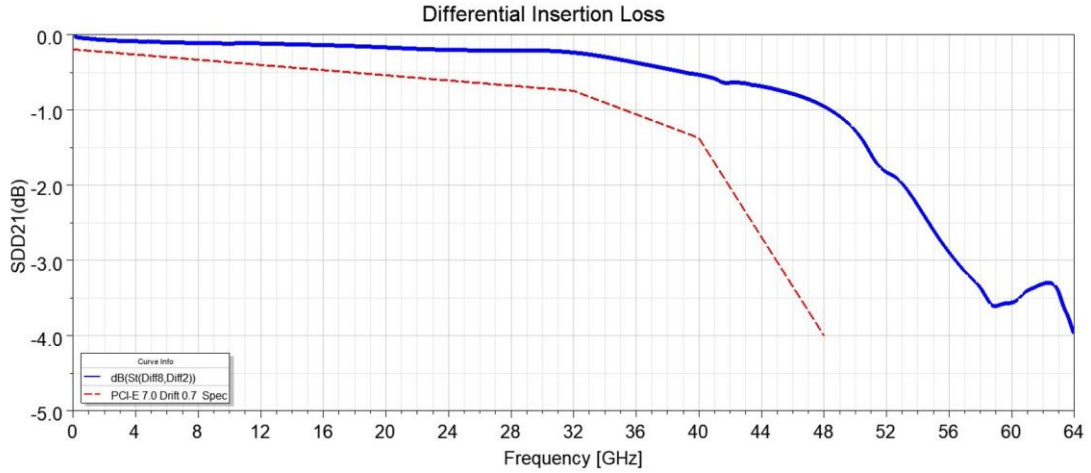




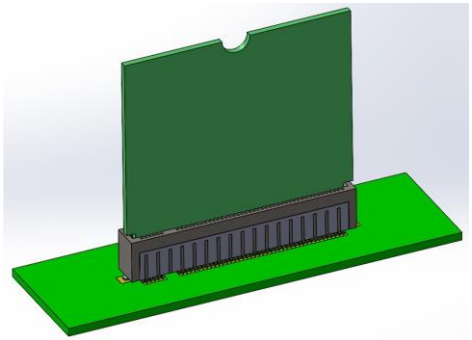
OC-DDIMM Connector, M.2 Derivative, Design & PCIe-G7 SI

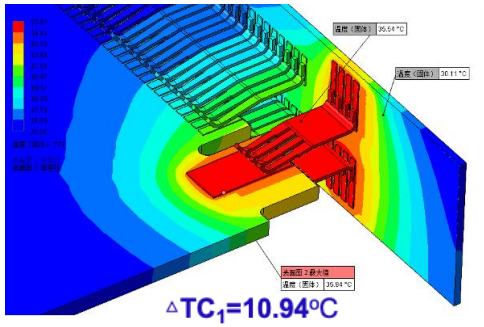


OC-DDIMM 96 Pin Connector

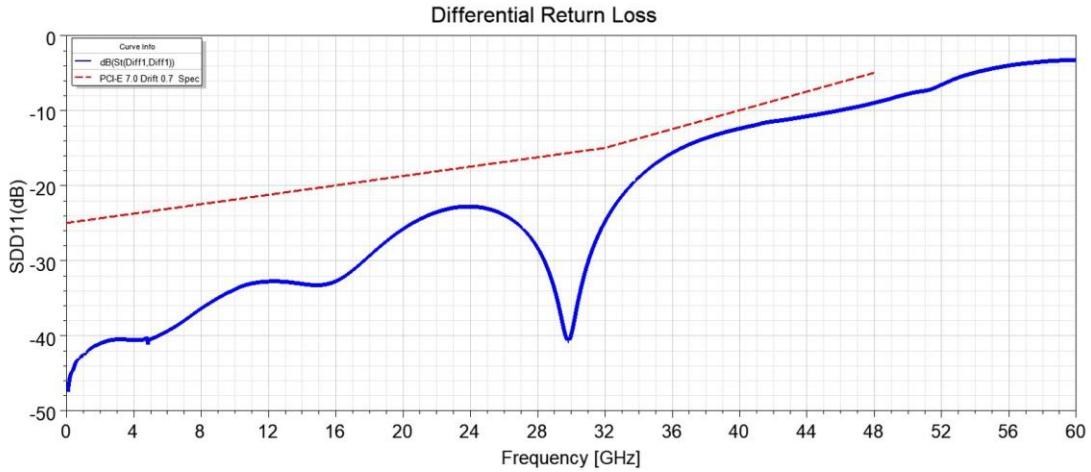


FIT belkin One Mobility **Amphenol**



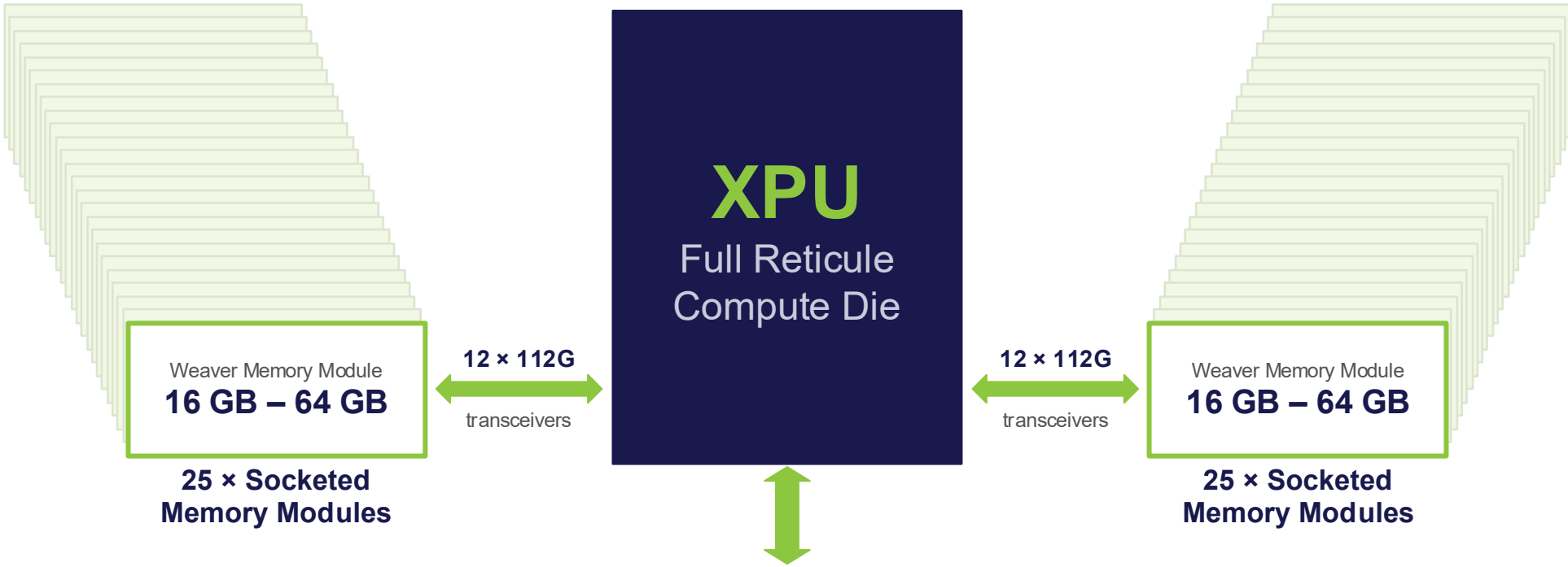


Current In : 1A per Pin @3.3V
Total 6 Pins supporting 20W



OCP HPCM Module — Block Diagram

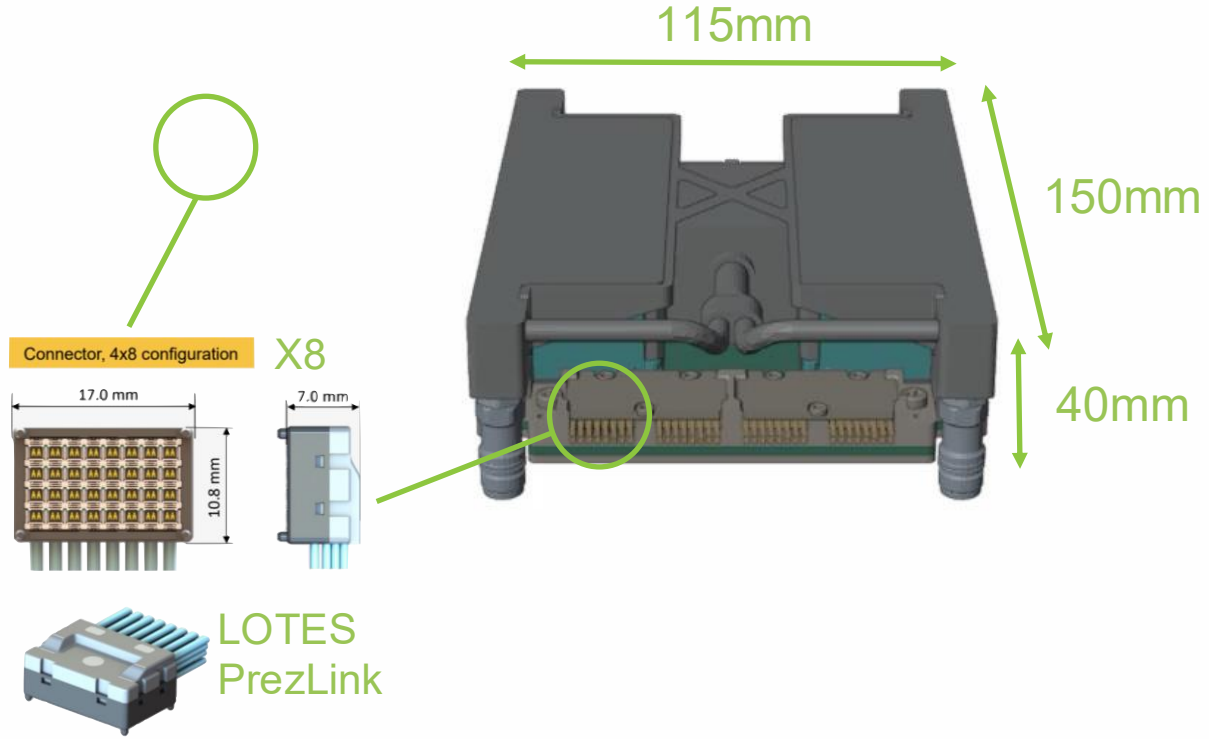
50 buffered-LPDDR memory modules over 600 × 112G OmniConnect lanes · 128 × fabric lanes



Front module shown in full; 24 identical modules staggered behind it on each side. 12 × 112G per module → 300 lanes per side, 600 lanes total. Capacity 0.8 TB – 3.2 TB per XPU module.

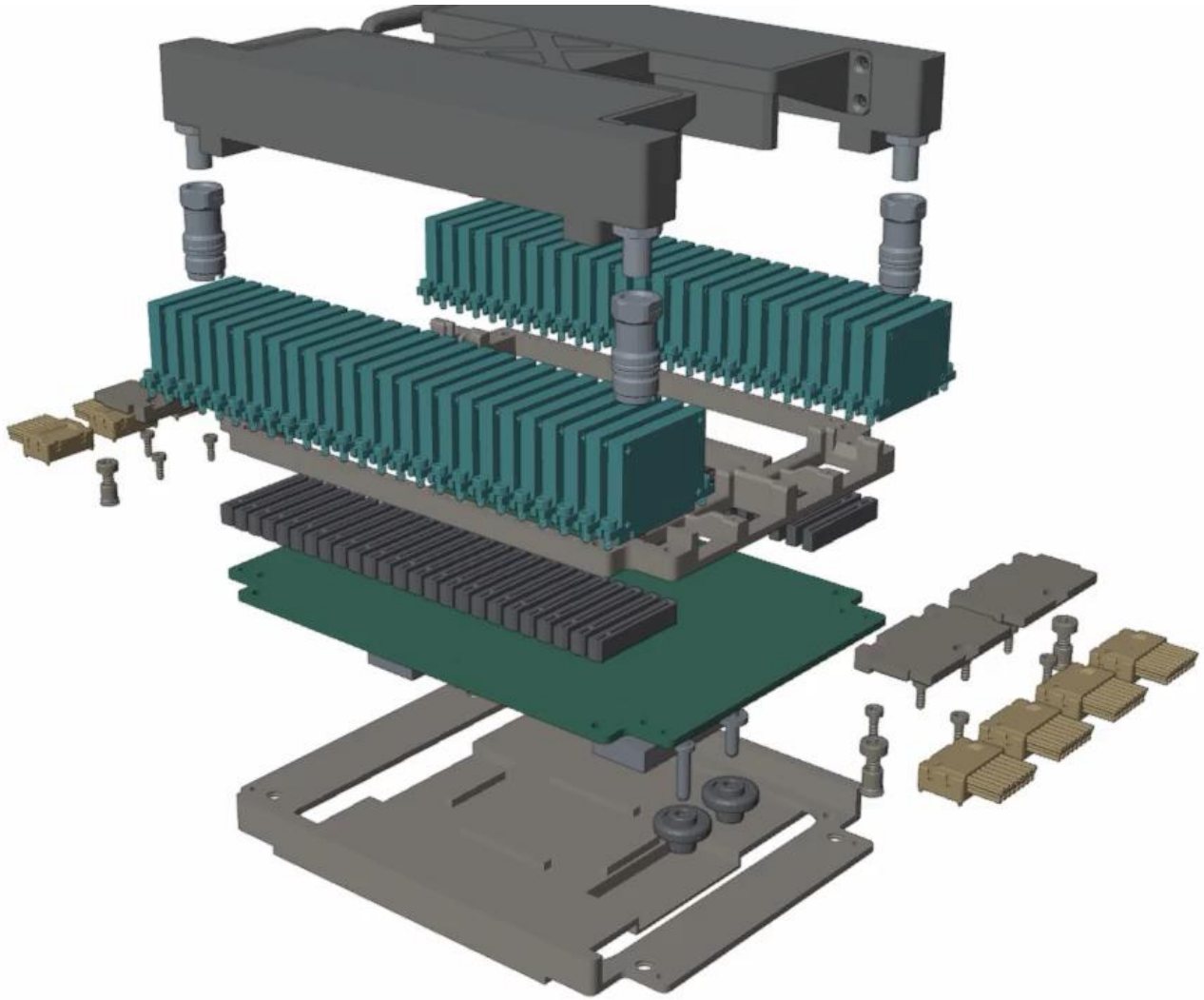
OCP HPCM 2026 Module - 50 Memory Modules + CPC IO

- Features
  - Standalone HPC XPU Module
    - No Motherboard required
  - 128 SerDes Co-Packaged Copper IO
    - LOTES PrezLink
  - Organic SubStrate with Monolithic Die
  - Hybrid Single / 2 Phase Water Cooled
  - Socketed CoPackaged Memory DIMMs
    - 50 Weaver Memory Modules
  - 48V Power Input, 1,650W TDP



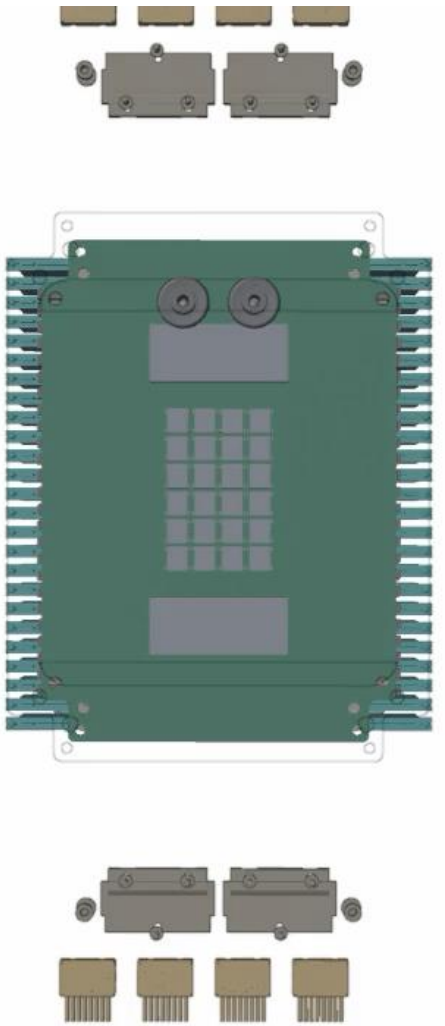
OCP HPCM 2026 Module - 50 Memory Modules + CPC IO

- Vertical Backside Power Delivery
  - LuxShare
    - 2x Unregulated 48V to 3.3V Modules
    - 250A Output per Module
  - EMPOWER (now part of Analog Devices)
    - 24x EP7502MC 65A POL Modules
- EP7144 used on OC-DDIMM Modules



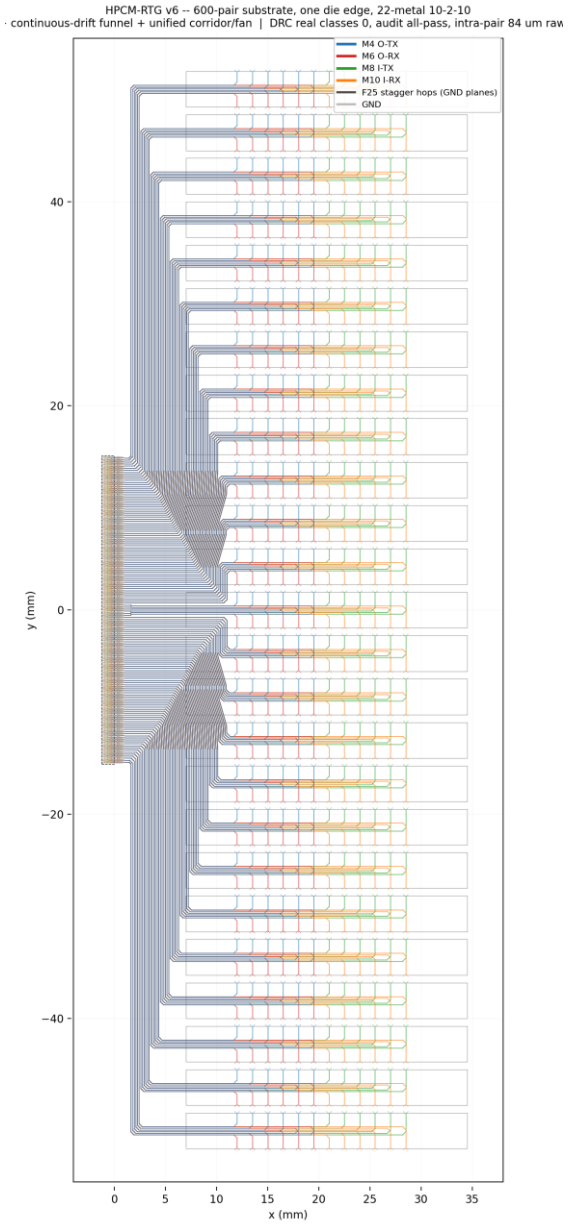
OCP HPCM 2026 Module - 50 Memory Modules + CPC IO

- Socketed CoPackaged Memory DIMMs
  - No Module Assembly Yield Loss
  - Serviceable
    - Replace Failing Memories
- All Interconnect On Substrate
  - 1.15pJ/bit for all XPU IO
  - Including between 18 XPUs in a 2OU Node



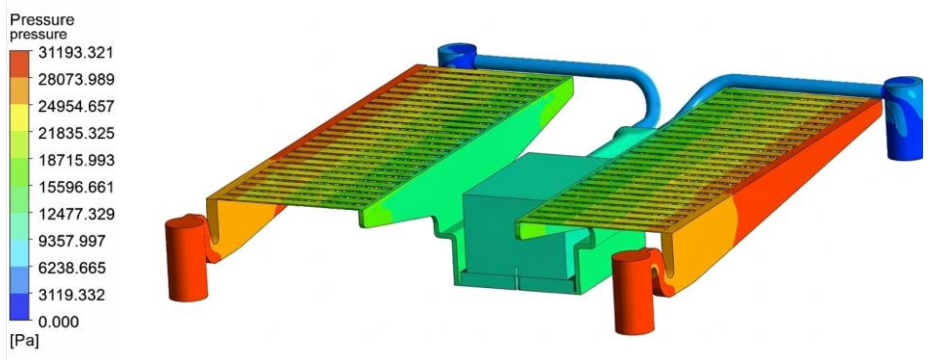
OC-DDIMM Routability Study

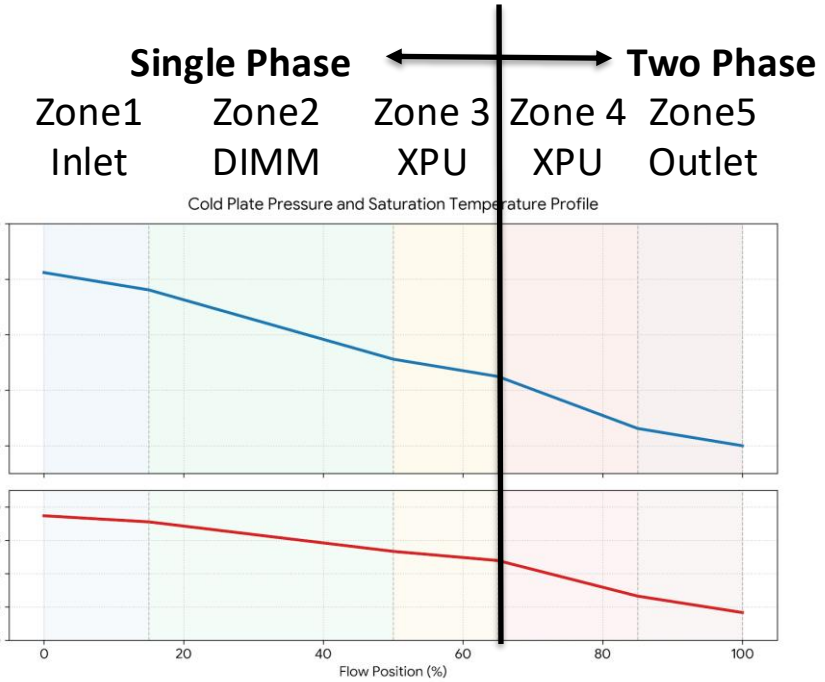
- 600 Diff Pairs routed from 33mm Reticule Die Edge to OC-DDIMMs
- 26um Trace Width
  - Relaxed for Large Substrate Yield
  - 15um Trace for Die Breakout
- 7.5dB Worst Case substrate routing Loss
  - XPU Die Bump to Connector Pad
- Total Channel OIF VSR Budget ~25dB?
  - Including Die Bump to Package Ball
  - Plenty of Margin

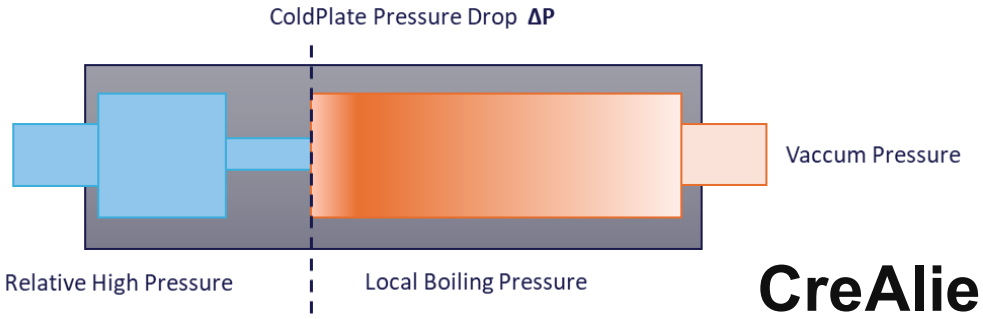


ColdPlate Design Architecture - with 2 Phase Water Cooling

- Hybrid Single/Two-Phase Architecture
  - Divides the cold plate into a Single-Phase Section (Zones 1–3) for DIMM preheating and a Two-Phase Section (Zones 4–5) for XPU latent heat dissipation.
- Cumulative Upstream Hydraulic Restrictor Strategy
  - The total cumulative single-phase pressure drop across Zones 1–3 acts as an integrated upstream throttling barrier. Suppress boiling back-pressure and preventing flow reversal.
- Sub-Atmospheric Thermal Clamping
  - Under an exit vacuum pressure of -90kPa, pure water boils at ~48C-53C, maintaining XPU junction temperatures in an optimal range.



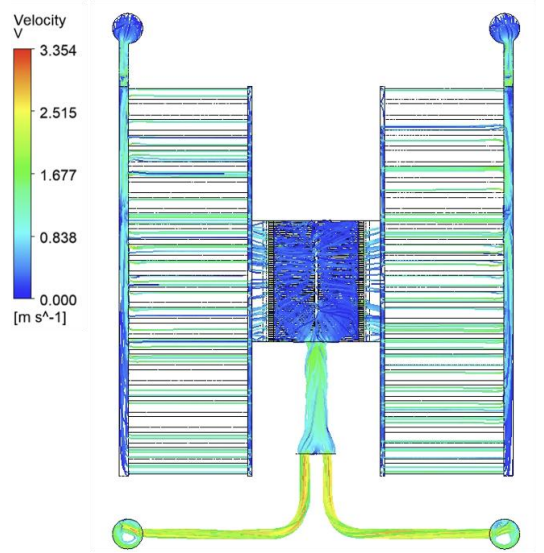


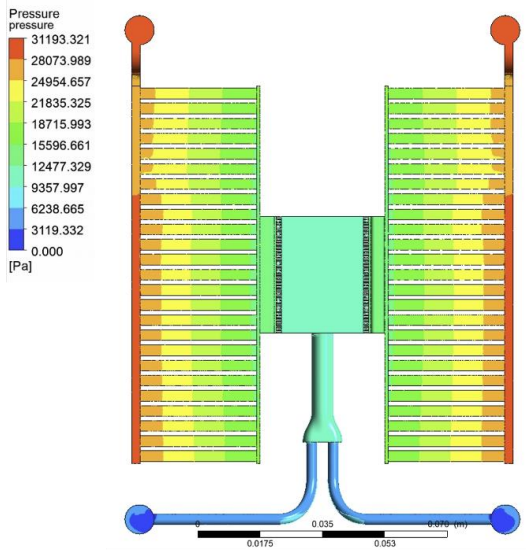


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Cold Plate Design

- CFD Flow Boundary & Power Capacity Target
  - Current CFD models a peak flow condition of 2 LPM total (1 LPM per side)
  - Based on empirical data (0.4 LPM / kW), this boundary evaluates hydraulic pressure limits for high-power cooling up to ~5 kW XPU load.
- Symmetric Flow Balancing & Uniform Distribution
  - Verifies balanced dual-inlet split flow across left and right DIMM micro-fin banks, ensuring uniform velocity and pressure drop before merging into the central core.
- Next Actions: Multi-Regime & Thermal Coupling Analysis
  - Low-Flow / Low-Power Dynamics: Evaluate system hydraulic behavior at reduced flow rates to analyze alternative flow regimes and stability limits.
  - Thermal Source Coupling: Integrate heat source inputs—prioritizing single-phase DIMM heat flux—to evaluate real-time fluid preheating and saturation point shifts.



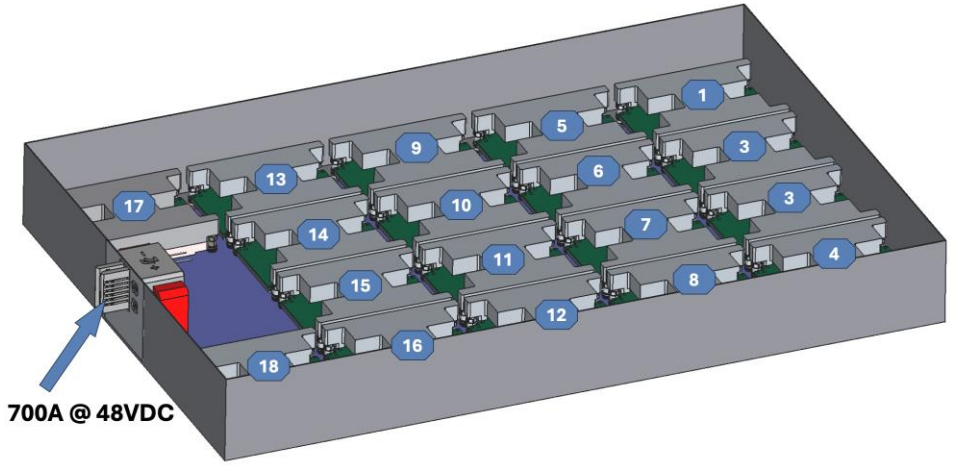


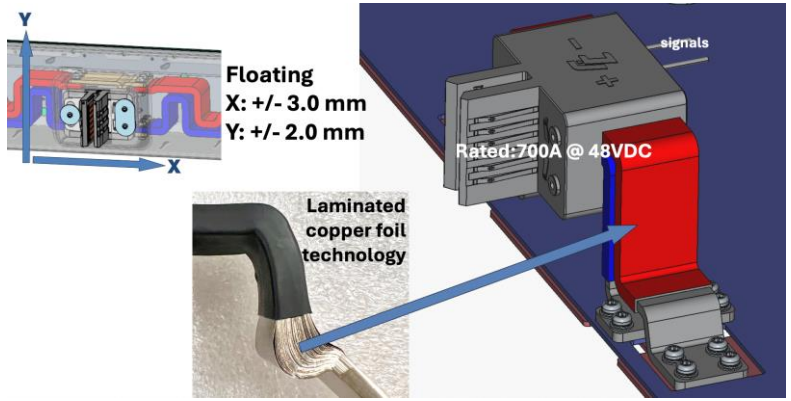


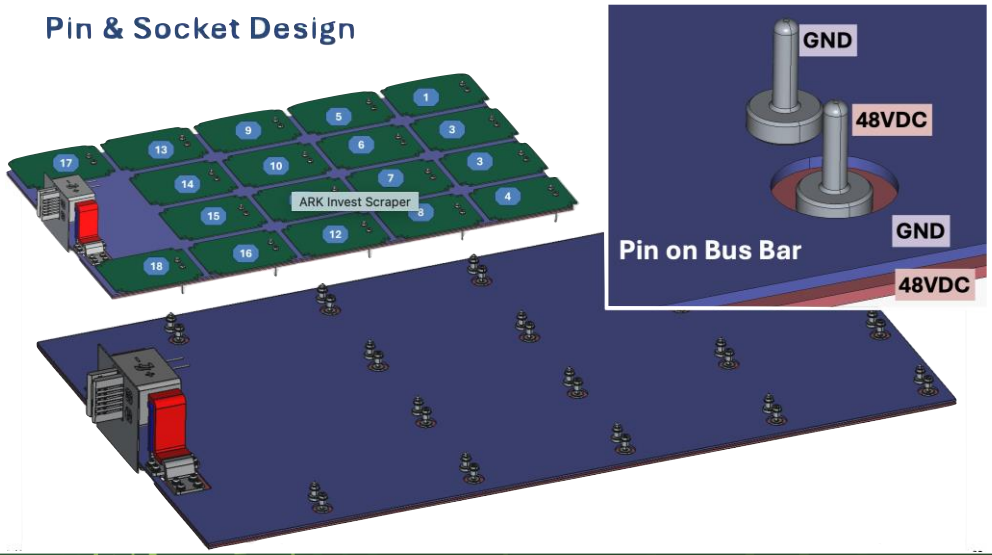
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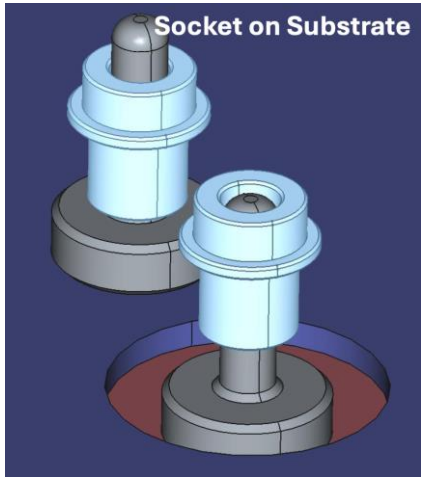
16 to 18 HPCM2026 Modules in an ORV3 2OU Chassis

- Power & Water Cooling provided from below Modules
- Modules Blind Mate into Power and Water
- 30KW Power Budget, 1.65KW per Module
  - 800W XPU
  - 850W Memory Modules









Call to Action - Help us bring this HPCM2026 Concept to POC

- We need help to bring this concept to reality
  - Especially adopters & Interested parties in Credo's OmniConnect Technology
  - Help create a compelling PoC that can grow into a competitive viable ecosystem
- Join our OCP HPC Subproject Workgroup
  - Mailing list: https://ocp-all.groups.io/g/OCP-HPC
  - Wiki : https://www.opencompute.org/wiki/HPC
  - Meeting Calendar : https://www.opencompute.org/projects/project-and-ic-meetings-calendar
  - Every other Tuesday, 8am Pacific

Thank You!

